

ABSTRACT

Malaria remains a serious health threat in Sub-Saharan Africa, particularly for pregnant women and young children. In countries such as Nigeria, the widespread circulation of falsified and substandard antimalarial medicines continues to undermine public health efforts. These challenges reflect more than just regulatory or logistical failures; they result in deeply personal and often preventable harm. Vulnerable individuals may receive medicines that are ineffective or unsafe, leading to treatment failure, worsening illness, and, in many cases, avoidable death. Although progress has been made in malaria prevention and treatment, weak oversight, fragmented supply chains, and limited access to quality healthcare allow poor-quality medicines to reach those who are most at risk.

The study adopts a structured literature review methodology, guided by the World Health Organization's Health Systems Framework, to examine how systemic health system weaknesses contribute to the persistence of this issue. Sources were selected from academic databases and global health organizations, with a focus on studies published between 2000 and 2025. Key themes were identified, including the prevalence of poor-quality medicines, health impacts on mothers and children, regulatory and supply chain challenges, and current policy responses.

Findings reveal that inadequate oversight, stock shortages, and reliance on informal markets continue to undermine malaria control efforts. In contrast, Rwanda demonstrates that strong regulatory governance, community-based care, and investment in health infrastructure can reduce the burden of falsified medicines. This research underscores the urgent need for collaboration across sectors to build stronger health systems, maintain the quality of medicines, and make effective malaria treatment accessible to everyone. By focusing on the real-life experiences of those most impacted by the disease, the study brings attention to the critical role that trust, safety, and accountability play in creating public health systems people can rely on.

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INTRODUCTION

Malaria remains one of the most serious health challenges in Sub-Saharan Africa, where most of the world's cases and deaths still occur (World Health Organization, 2024). Pregnant women and young children are particularly at risk because of their biological vulnerability and difficulties in accessing timely and effective healthcare. In Uganda, malaria is the leading cause of death among children under five and is closely linked to problems such as low birth weight and premature births (Uganda Bureau of Statistics, 2018).

While there have been advances in diagnosing, preventing, and treating malaria around the world, this progress is now threatened. The spread of falsified and substandard antimalarial medicines weakens treatment efforts and puts the most vulnerable people in the region at even greater risk. Sub-Saharan Africa continues to carry a heavy burden of malaria. In 2020, children under five made up most of the malaria-related deaths in this area (Oladipo, 2022). Recently, malaria control has faced setbacks due to growing resistance to common insecticides, the development of partial resistance to key treatments, and ongoing funding shortages (Oladipo, 2022). Beyond health, malaria also slows economic growth by over one percent each year in affected countries and creates high costs through treatment expenses, lost productivity, and strain on weak health systems (Keiser, 2004).

Health systems that are already fragile make the problem worse. Many places regularly experience shortages of medicines and diagnostic tools, have limited ways to track the disease, and lack enough trained healthcare workers. Poverty, poor living conditions, and low health knowledge increase risks, especially in rural areas where distance, costs, and mistrust often delay people from seeking care or push them to use informal providers. The impact is most severe for pregnant women and young children, who face higher chances of anemia, low birth weight, premature birth, and death (Uganda Bureau of Statistics, 2018).

Environmental and social factors also contribute to how malaria spreads. The tropical climate with warm temperatures and heavy rainfall creates perfect breeding conditions for mosquitoes, leading to year-round transmission. Rapid urban growth makes this worse, especially when it leads to informal settlements with standing water and poor waste management, which are ideal for mosquito breeding (Neiderud, 2015). While some countries like Rwanda and Botswana have made progress through strong national programs and community involvement, others, including Nigeria and the Democratic Republic of Congo, still have high rates of malaria and death. This shows how uneven the malaria burden and health system strength are across the region.

Medicine quality is another urgent issue. About one in ten medical products in low- and middle-income countries do not meet quality standards, meaning they may be falsified or substandard (World Health Organization, 2024). These medicines might lack the active ingredients, contain the wrong substances, or even be contaminated, all of which threaten patient safety and contribute to drug resistance (Newton, 2010). Rural communities are especially vulnerable because they often fall outside the reach of regulators and essential health services (Wada, 2022). Tackling poor-quality medicines is complicated by many factors. These include limited funding and workforce, weak regulatory systems, lack of price controls, and insufficient healthcare technology. Natural disasters, conflicts, health emergencies, supply chain problems, poor management, and poverty make these problems worse (Wada, 2022).

Globally, measures like the European Union's 2019 Falsified Medicines Directive require strict anti-counterfeiting steps such as serialization and tamper-proof packaging (Wada, 2022). However, many African countries only require basic labeling, like batch numbers and expiry dates, according to manufacturing guidelines (Wada, 2022). The heavy reliance on out-of-pocket healthcare payments drives many people to informal markets where they risk buying poor-quality medicines (Wada, 2022). In Uganda, medicine supply works through a system where hospitals and health centers send requests every two months to the National Medical Stores. Each facility should have a committee responsible for planning, storing, and distributing medicines. Still, poor coordination among healthcare workers, government, donors, and local leaders makes

it difficult to maintain a steady supply (Yadav, 2015). The production of counterfeit drugs is mostly driven by profit, and those responsible often face few legal consequences. These fake medicines usually target diseases like malaria, which mainly affect marginalized and low-income groups (Cockburn, 2005).

METHODOLOGY

This study is based on a review of existing research and data that looks at how falsified and substandard antimalarial medicines affect the health of mothers and children in Sub-Saharan Africa. Because this issue spans many countries and involves complex systems, reviewing published literature was the most practical way to understand the broader patterns and identify gaps in policies and practices.

The World Health Organization's Health Systems Framework served as a guide throughout this work. This framework highlights six important parts of a strong health system: how services are delivered, the health workforce, health information systems, access to essential medicines, health financing, and leadership and governance (World Health Organization, 2010). Using this framework helped organize the analysis by showing how problems in these areas can lead to poor-quality medicines reaching vulnerable populations.

To find relevant information, academic databases and global health organizations were searched between March and July 2025. These included PubMed, Google Scholar, the WHO Global Observatory, and reports from the World Bank, UNICEF, and the Global Fund. Searches combined keywords such as "substandard antimalarial medicines," "malaria drug quality Africa," "maternal health malaria," and "pharmaceutical regulation Sub-Saharan Africa." Only sources published in English between 2000 and 2025 were included to keep the review current and accessible.

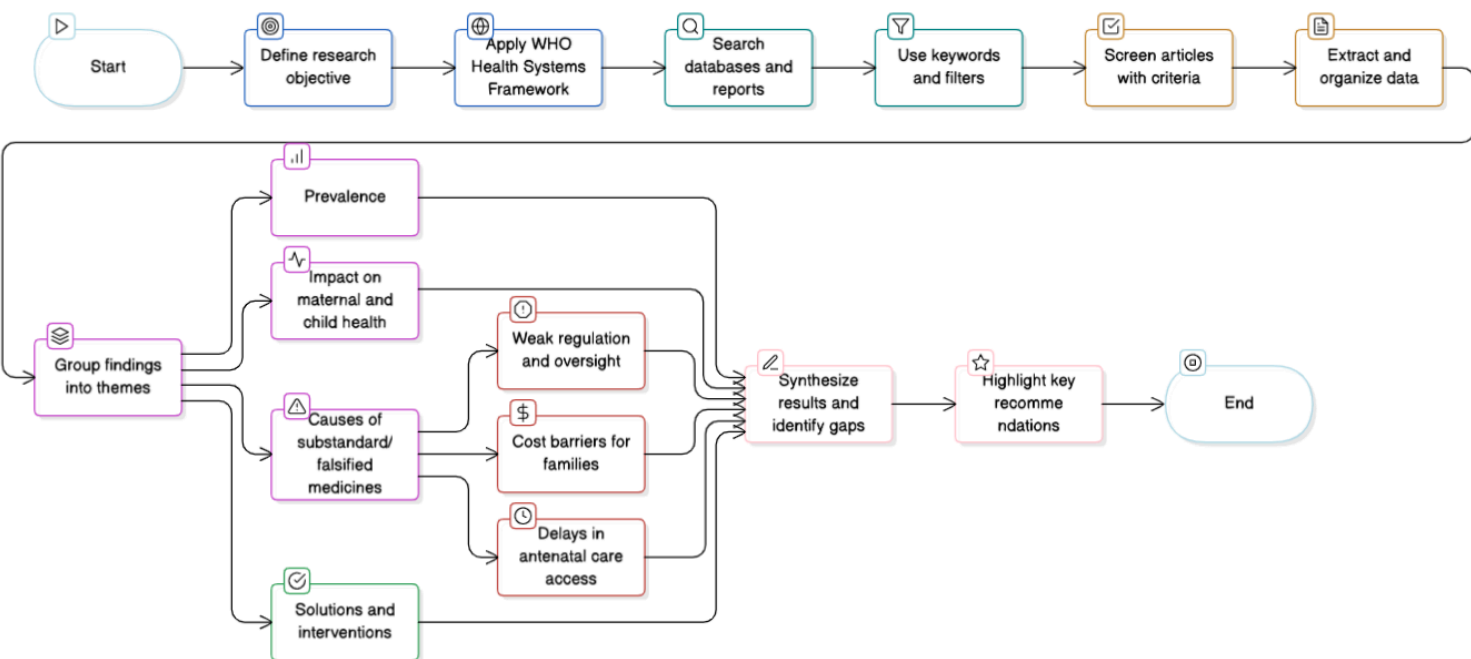


Figure 1. Flowchart summarizing the methodology and literature review process used in this study.

The focus was on studies that examined the quality of antimalarial medicines, the distribution of medicines, regulatory systems, and health outcomes for pregnant women and children in Sub-Saharan Africa. Articles that only examined drug chemistry, development, or non-African regions were excluded. Priority was given to peer-reviewed studies and reports from trusted organizations to guarantee the reliability of the information. Information from the selected sources was carefully reviewed and grouped into four main themes. These were the extent of falsified and substandard medicines in the region, their impact on mothers' and children's health, the underlying causes, such as weak regulation and supply chain problems, and possible solutions like policy changes and strengthening health systems.

Because this study used only existing data, no ethical approval was necessary. However, there are some limitations. The review might miss the most recent changes that have not yet been published, and by only including English-language sources, research from French or Portuguese-speaking countries could have been left out. There

is also a chance of publication bias since studies with unclear or negative results may not be as widely published. Lastly, without collecting new data, this study cannot reflect the personal experiences of patients or healthcare workers, which are important for understanding the real challenges people face. Despite these limitations, this review offers important insights into how poor-quality antimalarial medicines continue to affect maternal and child health. It highlights weaknesses in health systems and regulations, pointing toward areas where better policies, funding, and leadership could make a real difference. This work is intended to support future research and guide policymakers and health leaders in improving medicine quality across Sub-Saharan Africa (Figure 1).

RESULTS

Theme 1: Prevalence and Nature of Poor-Quality Antimalarials

Between 2013 and 2017, nearly half of all reported cases of falsified medicines worldwide were reported in Africa (World Health Organization, 2024). During this period, about half of the medicines sold online were found to be counterfeit, posing a serious global health threat (World Health Organization, 2024; Newton, 2010). The situation is even more concerning when focusing specifically on antimalarial drugs. Studies conducted across 21 countries in sub-Saharan Africa between 2001 and 2010 found that more than one-third of antimalarial medicines tested failed to meet quality standards (Renschler et al., 2016; Nayyar, 2012). This issue is particularly widespread in rural areas and informal markets, where regulation and quality control tend to be weakest (Nayyar, 2012).

Nigeria faces one of the most severe parts of this crisis. It distributed approximately 30 million courses of antimalarial treatment to children under five during this period, but nearly two-thirds of these medicines were of poor quality. Unsafe medicines are estimated to cause more than 12,000 preventable deaths annually, mostly among young children (Mekonnen, 2024; Issa, 2023). As a result, Nigeria has the highest estimated number of child deaths linked to treatment failure caused by substandard medicines (Renschler, 2016; Beargie, 2019). Weak regulatory systems and poor management of medicine supply chains are the main drivers of this problem, common in many low-income countries (Vledder, 2019).

Globally, the World Health Organization estimates that half of the world's population does not have consistent access to essential medicines; in sub-Saharan Africa, the situation is more severe, with about two-thirds lacking consistent access (World Health Organization, 2024). The presence of falsified and substandard medicines exacerbates this problem. Research indicates that approximately 17 percent of antimalarial medicines contain incorrect doses, and 16 percent have entirely wrong ingredients. These risks are highest when medicines are purchased from private pharmacies or informal sellers (Tipke, 2008). Not only do poor-quality medicines fail to

effectively treat malaria, but they also contribute to the rise of drug-resistant strains, making future treatment more difficult (Newton, 2010). Many people in sub-Saharan Africa pay for healthcare out of pocket, which often leads them to purchase medicines from informal markets where the risk of encountering counterfeit drugs is higher (Wada, 2022). Addressing this issue requires stronger regulation, improved supply chain management, and increased public awareness to ensure that high-quality antimalarial medicines reach those who need them most.

Theme 2: Impact on Maternal and Child Health

Malaria during pregnancy remains one of the most urgent public health challenges in sub-Saharan Africa, with severe consequences for both mothers and newborns. It is closely linked to complications such as maternal anemia, stillbirth, preterm delivery, and low birth weight, all of which contribute to higher rates of neonatal mortality (Aberesse, 2022). These risks increase significantly when pregnant women do not have timely access to effective prevention and treatment. Studies show that inadequate care during pregnancy can lead to longer hospital stays and, in severe cases, maternal death (Aberesse, 2022).

Socioeconomic factors play a major role in shaping access to malaria prevention tools. Many women from low-income households, especially those lacking financial or emotional support from partners or family, face difficulties affording transportation to healthcare facilities. This limits their ability to obtain vital interventions like long-lasting insecticidal nets or malaria protection vouchers (Aberesse, 2022). Consequently, these women remain particularly vulnerable to malaria and its harmful effects throughout pregnancy.

The earliest stages of pregnancy carry unique risks. Malaria infections affect about one in five pregnant women in sub-Saharan Africa during the first trimester. While early infections may not directly cause low birth weight or poor neonatal outcomes, they often lead to maternal anemia later on and increase the chance of repeated infections during the second and third trimesters (Accrombessi, 2017). Research suggests that the malaria parasite can begin accumulating in the placenta as early as eight weeks into

pregnancy, disrupting its development and function, which can ultimately harm the baby's growth and health (Chua et al., 2021). Additional evidence links early malaria infections to hypertensive disorders such as preeclampsia, especially in first-time mothers. These findings indicate the critical need for early intervention to reduce risks before they worsen later in pregnancy.

Even though clear guidelines exist, many pregnant women in sub-Saharan Africa begin antenatal care too late to take full advantage of preventive malaria treatments. The World Health Organization advises that pregnant women should receive at least three doses of sulfadoxine-pyrimethamine for malaria prevention, starting in their second trimester. However, late attendance and frequent drug shortages mean that many women receive fewer than the recommended doses or none at all (Aberesse, 2022). This is especially concerning given evidence showing that three or more doses significantly lower the risk of low birth weight and stillbirth, even in areas where some resistance to the drug has been reported (Awoke, 2025).

In addition to individual barriers, broader structural and system-wide challenges limit the effectiveness of malaria prevention. Many countries struggle to accurately track coverage of intermittent preventive treatment due to weak health information systems. Much of the data comes from periodic national surveys rather than continuous clinic reporting, making it difficult to identify and address gaps in coverage (Wen, 2016).

Although most countries in the region have adopted WHO-recommended strategies for preventing malaria during pregnancy, including early diagnosis, access to insecticide-treated bed nets, and preventive medication implementation remains uneven. In many rural and low-income areas, pregnant women are often unable to access these services because of poor infrastructure, delays in starting antenatal care, or frequent shortages of essential drugs and supplies. These challenges expose both mothers and their babies to unnecessary health risks and highlight the urgent need for more focused efforts to ensure malaria prevention reaches every woman, regardless of her location.

Theme 3: Supply Chain and Regulatory System Weakness

Sub-Saharan Africa continues to face significant challenges in regulating and maintaining the quality of medicines, which have serious and far-reaching impacts on public health. Between 2013 and 2017, the World Health Organization reported that 42 percent of all detected falsified medicines worldwide originated in Africa. Even more concerning, half of all drugs sold online during that period were found to be counterfeit (Wada, 2022). These poor-quality medicines contribute to devastating health outcomes, causing an estimated one million deaths annually. This number includes roughly 450,000 malaria-related deaths that could have been prevented (Wada, 2022).

In many African countries, a large share of medicine distribution occurs through informal or loosely regulated channels such as local patent medicine vendors and open markets. Often, these sellers operate outside legal boundaries and lack proper training or government oversight to ensure the safety of the medicines they provide. In some regions, the only requirement to open a medicine shop is basic literacy, which is inadequate given the potential health risks involved (Wada, 2022). This situation has created a fragmented and poorly controlled pharmaceutical market where counterfeit and substandard medicines are widespread, especially among street vendors and unregulated sellers. Despite being illegal, street hawking of medicines remains common across the region.

Although many African countries have policies aimed at regulating medicine quality, only about 15 percent have the legal authority and capacity to fully enforce these regulations (Wada, 2022). For example, the European Union introduced strict laws in 2019 requiring prescription drugs to have serialized tracking and tamper-proof packaging to prevent counterfeit medicines from entering the market (Wada, 2022). In contrast, most African countries still rely on basic labels like batch numbers and expiration dates, which are insufficient to prevent falsification (Wada, 2022). To address these challenges, African Union member states launched the African Medicines Agency in 2019. This continental initiative seeks to harmonize medicine regulations and improve oversight across countries. While this is a promising step forward, much work remains to strengthen supply chains, enforce regulations effectively, and train staff to assure medicines are safe and effective (Ncube, 2021).

Nigeria faces particularly significant challenges in improving the quality of antimalarial medicines, despite ongoing efforts. Studies indicate that eliminating stock shortages of malaria drugs nationwide could save over \$1.1 billion each year (Beargie, 2019). Removing fake and substandard antimalarials could also save approximately \$892 million annually. However, focusing improvements only on artemisinin-based combination therapies would yield smaller savings of around \$161 million per year. This highlights the complexity of Nigeria's malaria treatment environment, where patients depend on a wide range of medicines beyond just ACTs (Beargie, 2019).

The structure of Nigeria's pharmaceutical market plays a key role in the continued circulation of poor-quality malaria medicines. Although the National Agency for Food and Drug Administration and Control has made efforts to regulate supply chains and improve drug quality, challenges remain. The distribution system is fragmented, local manufacturing is limited, and regulatory processes face numerous obstacles. These weaknesses create opportunities for unethical practices such as bribery and diversion of donated medicines (Beargie, 2019; Onwujekwe, 2018). Nigeria also relies heavily on medicine imports, especially from countries like India and China, where some substandard products have been identified, further complicating quality assurance efforts (Onwujekwe, 2018).

Environmental and logistical factors also degrade medicine quality. Poor storage and transport conditions, particularly in hot and humid areas, can reduce the effectiveness of medicines (Nayyar, 2012). Distribution challenges mean that medicines often expire before reaching patients, contributing to the spread of ineffective drugs (Beargie, 2019). This situation shows just how important it is to better manage medicine supply chains and provide proper storage to preserve medicine quality.

To tackle these issues, Nigeria must strengthen its regulatory system by empowering NAFDAC to better license manufacturers, enforce good manufacturing practices, and conduct thorough quality inspections (Onwujekwe, 2018). Closer collaboration between NAFDAC and the Ministry of Health could enhance medicine distribution, storage, and inspection processes (Ncube, 2021). These reforms are essential not only to protect

patients from fake or degraded medicines but also to rebuild trust in the healthcare system and ensure that malaria treatments are effective.

Stock shortages remain a persistent problem in Nigeria's antimalarial supply chain, affecting both public and private health facilities. These shortages stem from limited funding, a lack of trained personnel for medicine procurement, and weak storage and distribution systems (Beargie, 2019; Onwujekwe, 2018). As a result, many patients turn to informal sources such as street vendors and unregulated chemists, where the risk of obtaining counterfeit or poor-quality medicines is much higher (Beargie, 2019).

Most informal sellers lack formal pharmaceutical training, limiting their ability to guarantee medicine safety. Improving inventory management and training more licensed pharmacists in both the public and private sectors could help reduce shortages and decrease patients' reliance on unregulated markets (Onwujekwe, 2018). Research suggests that addressing shortages in public health facilities alone could save Nigeria's health system about \$732 million annually, with an additional \$416 million in savings possible if the private sector also improves (Beargie, 2019). These figures demonstrate the importance of addressing these challenges to safeguard public health and promote economic stability.

Theme 4: Interventions and Policy Strategies

Sub-Saharan Africa continues to face ongoing challenges in regulating and assuring the quality of pharmaceutical supply chains, with serious consequences for public health. Between 2013 and 2017, the World Health Organization reported that 42 percent of all falsified medicines detected worldwide originated from Africa. Even more concerning, half of the drugs sold online during this period were identified as counterfeit (Wada, 2022).

In many African countries, a large portion of medicine distribution happens through informal and weakly regulated channels such as Patent and Proprietary Medicine Vendors, Patent Medicine Sellers, and open drug markets. Many of these vendors operate beyond their legal scope, selling medicines without adequate training

or oversight. In some areas, the only requirement to run a patent medicine shop is basic literacy, which raises serious concerns given the risks of improper dispensing (Wada, 2022). This situation has created a fragmented pharmaceutical landscape where counterfeit and substandard medicines are widespread, especially in unauthorized markets and street vendors. Although street hawking of medicines is illegal, it remains common across the region.

While formal policies for medicine regulation exist in many African countries, only about 15 percent have the legal authority and capacity to enforce all core regulatory functions effectively (Ncube, 2021). In contrast, the European Union introduced strict anti-counterfeiting regulations in 2019 through its Falsified Medicine Directive, which requires serialization and tamper-evident packaging for prescription drugs. Most African nations continue to rely mainly on basic labeling, such as batch numbers and expiration dates, which do not sufficiently prevent falsification (Wada, 2022). To address these challenges, African Union member states agreed in 2019 to establish the African Medicines Agency, aiming to harmonize medicine regulation across the continent. While this agency marks important progress, much work remains to strengthen enforcement, training, and supply chain governance (Ncube, 2021).

Nigeria faces significant hurdles in improving the quality of antimalarial medicines despite various interventions. Studies suggest that eliminating stock-outs of malaria drugs nationwide is the most cost-effective strategy, potentially saving over \$1.1 billion annually (Beargie, 2019). Removing substandard and falsified antimalarials could also save nearly \$900 million each year. However, focusing improvements solely on artemisinin-based combination therapies without addressing other commonly used treatments would result in more limited savings of about \$161 million annually (Beargie, 2019). This reveals just how complicated malaria treatment is in Nigeria, where patients rely on a wide range of medicines, not only ACTs, to fight the disease.

The structure of Nigeria's pharmaceutical market contributes significantly to the ongoing circulation of poor-quality antimalarials. Although the National Agency for Food and Drug Administration and Control (NAFDAC) has worked to regulate supply chains

and improve drug quality, these efforts have not eliminated the widespread availability of substandard products. The sector is characterized by fragmented distribution systems, limited local manufacturing, and regulatory challenges, creating opportunities for unethical practices such as bribery and diversion of medicines (Onwujekwe, 2018). With few pharmaceutical manufacturers in the country, to address these challenges, Nigeria needs to strengthen its pharmaceutical regulatory framework and build capacity to license manufacturers, enforce good manufacturing practices, and conduct rigorous quality assessments. Improved coordination between NAFDAC and the Federal Ministry of Health could help close gaps in distribution, storage, and inspection processes. These reforms are critical for protecting patients from falsified or degraded medicines and for maintaining trust in the health system (Ncube, 2021).

The pharmaceutical supply chain for antimalarial medicines in Nigeria also suffers from frequent stock-outs in both public and private healthcare facilities. These shortages largely result from systemic inefficiencies such as limited funding, insufficient training of procurement personnel, and weak logistics management (Yadav, 2015). Consequently, many patients turn to informal sources such as drug hawkers or unregulated chemists, where substandard and falsified medicines are more common (Wada, 2022). These informal vendors often lack formal pharmaceutical training, which limits their ability to ensure medicine safety and quality. Improving inventory management systems and building the capacity of licensed pharmacists in both sectors could reduce stock-outs and lessen patients' reliance on unregulated markets (Yadav, 2015). Research also shows that eliminating stock-outs in public health facilities alone could save Nigeria's health system approximately \$732 million annually, with another \$416 million saved through improvements in the private sector (Beargie, 2019).

DISCUSSION

This study takes a closer look at the persistent problem of poor-quality antimalarial medicines in Sub-Saharan Africa, with a special focus on Nigeria and important insights from Rwanda. It dives into four main areas: how common the problem is, the effects it has on mothers and children, the gaps in regulation and supply chains, and the different ways countries are working to tackle these challenges. Altogether, these themes highlight just how complicated and serious the issue is. But they also offer hope, showing that meaningful progress can be made when the right approaches are put into action.

There is strong evidence that substandard and falsified antimalarial medicines continue to circulate widely across the region. Between 2013 and 2017, nearly half of all reported cases of falsified medicines globally came from sub-Saharan Africa (World Health Organization, 2024). In Nigeria, the scale of the problem is especially concerning. More than 30 million malaria treatments were given to children under five during this period, and almost two-thirds of those medicines were found to be of poor quality (Wada, 2022). When medications are weak or fake, they often fail to treat the disease properly, leading to worsening illness or even death. Over time, this also contributes to drug resistance, making it harder to treat malaria effectively in the future.

These risks are even more troubling when it comes to maternal and child health. Pregnant women and young children are more likely to experience severe outcomes from malaria. Infections during pregnancy have been linked to anemia in mothers, stillbirths, premature deliveries, and low birth weight in newborns. This study found that around one in five women experience malaria early in pregnancy, and while these early infections may not always cause immediate complications, they often lead to later infections that carry greater risks. The World Health Organization recommends that pregnant women receive at least three doses of preventive malaria treatment, but in Nigeria, many do not complete the full course. Delays in starting antenatal care, frequent medicine stock-outs, and difficulties accessing health facilities, especially in rural and poorer areas, all contribute to this gap.

Behind these health threats are deeper problems within the systems responsible for regulating and distributing medicines. In many parts of Nigeria, people often buy antimalarial drugs from informal sources like street vendors or open markets. These vendors typically lack formal medical training, and there is little to no oversight of the medicines they sell. As a result, patients may receive expired, counterfeit, or improperly stored drugs. Although it is illegal to sell medicines this way in Nigeria, weak enforcement of the law means these practices continue to be widespread.

Another significant challenge is the limited capacity of many African countries to monitor and regulate medicine quality effectively. National agencies often lack the resources and authority to inspect imported shipments, maintain proper storage conditions, or test whether medicines meet required standards. In Nigeria, medicine stock-outs remain common in both public and private clinics. Even when medicines are sourced through official channels, they can lose effectiveness due to heat or humidity during storage and transport. Nigeria's heavy dependence on imported medicines, particularly from India and China, makes it harder to maintain control over product quality. Poor governance further compounds these problems. Corruption, limited accountability, and a lack of coordination between key agencies such as customs, procurement offices, and health ministries create vulnerabilities across the entire system. Regulatory bodies may not be truly independent and can be subject to political influence, weakening their ability to enforce quality standards.

In contrast, Rwanda offers a promising example of what is possible when a country strengthens its systems. The Rwandan government has taken clear steps to improve regulation and oversight. All imported medicines are inspected, and suppliers must meet international quality standards. Healthcare workers are trained to recognize and report suspicious medicines, creating an early warning system for potential problems. In one cross-country study, Rwanda was the only country where none of the tested antimalarial samples were found to be fake or of poor quality (Binagwaho, 2013). This suggests that a strong regulatory approach can make a real difference in protecting patients.

Rwanda's success also reflects its commitment to community-based healthcare. Community health workers are essential to malaria prevention, helping pregnant women receive preventive treatment, distributing insecticide-treated bed nets, and referring patients to clinics when needed. This people-centered approach brings care closer to where people live, improves access, and builds trust. Rwanda has also used mobile technology, such as SMS-based systems, to track medicine stocks and prevent stock-outs, helping ensure medicines are available when and where they are needed (Binagwaho, 2013).

Making antimalarial medicines more affordable is an important step, especially for families struggling with poverty. One way to lower prices is by removing taxes and import duties on these essential medicines. Additionally, setting limits on how much can be added to the price at each stage of the supply chain can help keep costs reasonable. Countries can also work together through pooled procurement, buying medicines in bulk to negotiate better deals. Investing in local production of medicines, streamlining how orders are placed, and expanding national health insurance coverage can ease the financial pressure on patients even more. Encouraging pharmaceutical companies to focus on creating high-quality, affordable treatments by offering incentives such as grants, technical support, or reliable purchasing agreements can help shift the market toward better, more accessible products.

While this study offers a valuable understanding, it does have some limitations. It is based entirely on existing literature and publicly available data, which means it might not reflect the very latest changes in policy or technology. Additionally, the study did not include interviews or field research, so it misses out on the firsthand perspectives of patients, health workers, and regulators who face these challenges every day.

From a policy perspective, there is a clear need for stronger regulation, more resilient supply chains, and greater investment in local pharmaceutical systems. Countries like Nigeria could benefit from improving enforcement, expanding public awareness, and increasing coordination among health and non-health agencies. Rwanda's experience shows that success is possible through thoughtful planning,

strong governance, and sustained commitment. Ultimately, poor-quality antimalarial medicines remain a serious threat in sub-Saharan Africa, particularly for women and children. Nigeria continues to face major challenges, but the example of Rwanda shows that with the right systems and leadership, countries can protect their populations and make meaningful progress. By investing in smarter, people-centered strategies, governments have the opportunity to strengthen their health systems and reduce the burden of malaria for the communities that need support the most.

CONCLUSION

The widespread circulation of substandard and falsified antimalarial medicines in sub-Saharan Africa is more than a technical issue. It is a deeply human one, with real consequences for people who are simply trying to protect themselves and their families from disease. At its core, this problem speaks to how health systems either succeed or fail in reaching those who need care the most. It reminds us that a medicine is only effective if it is trusted, accessible, and safe, and that no child, no mother, and no community should be left to navigate these challenges alone.

Throughout this research, one truth has become clear: poor-quality medicines are not just the result of weak regulation or broken supply chains. They are symptoms of larger structural problems that affect how people experience healthcare in everyday life. When pregnant women arrive late to antenatal visits because they live far from the nearest clinic, or when families turn to informal vendors because formal care is too expensive or too far away, it reflects deeper issues of inequality, geography, and trust. And when public health facilities themselves carry medicines that may be expired or ineffective, the very foundation of the health system is shaken.

The consequences are particularly severe for women and children, who remain at the center of the malaria burden. When medicine fails, the result is not only illness or death but also fear, uncertainty, and loss of faith in the systems that are meant to protect people. These outcomes cannot be addressed through policies alone. They require a shift in how healthcare is planned, funded, and delivered, with attention not just to what is being done but to how people experience it on the ground.

At the same time, this is not a story without hope. Rwanda's progress shows what can happen when leadership is committed, systems are well-organized, and community voices are valued. By investing in medicine inspection, empowering health workers, improving supply chains, and using simple technologies to monitor stock levels, Rwanda has built a system that not only delivers medicines but also builds confidence in care. This example is not about perfection. It reminds us that progress is within reach when the focus remains on people and the systems that serve them.

To move forward, Nigeria and other countries facing similar challenges must take a long-term view. Strengthening regulatory agencies is important, but it must be matched with better coordination between health ministries, customs, procurement teams, and local governments. Communities need to be engaged not only as beneficiaries but as active partners in designing solutions. Informal vendors, often painted as part of the problem, could instead be brought into the solution with training, licensing, and access to reliable supply chains. Investments in local pharmaceutical production must go hand in hand with strict quality control so that efforts to increase self-reliance do not lead to more harm.

Public health messages must also meet people where they are. Awareness campaigns, if done with respect and relevance, can help people make informed decisions and build trust in regulated healthcare providers. At the same time, affordability cannot be treated as a side issue. Health insurance, subsidies, and the removal of taxes on essential medicines are all critical in making quality care a realistic option for all families, not just a privileged few.

The issue of falsified and substandard medicines brings into focus the deeper questions of justice, equity, and dignity in global health. This research has shown that technical fixes, while necessary, are not enough. What is truly needed is a commitment to building systems that protect people not only from disease but from the hidden harms of poor-quality care. It means recognizing that every mother who walks to a clinic, every child who is given treatment, and every health worker who tries to do their best deserves to be supported by a system that works. As global health efforts continue to evolve, there is an opportunity to rethink how progress is measured. True success will not only be seen in reduced malaria cases but also in how many families feel safer, how many women feel heard, and how many children have access to care that is both effective and fair. The tools to solve this problem already exist. What remains is the collective will to act on them, and to do so in a way that centers the people whose lives are most at stake.

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